

Stairs, Handrails and Guards

The densest set of numbers in the whole exam, and they only make sense as one connected picture of a person climbing.

Every number here exists so a stair can be climbed in the dark without looking. That is why the tolerance between steps is so tight — the body learns the rhythm on the first step and repeats it blindly.

The numbers here

Riser — maximum height	7.75 in
Tread — minimum depth	10 in
Maximum variation between risers, or between treads	3/8 in
Nosing	3/4 to 1 1/4 in
Headroom	80 in (6 ft 8 in)
Clear width of a stairway	31 in
Maximum vertical rise between landings	12 ft
Stair landing — minimum dimension	36 in

Three-eighths of an inch is the one to hold onto. It is the same tolerance as the sidewalk trip hazard, and for the same reason: that is roughly the smallest step change a walking foot fails to notice.

Handrails

The numbers here

Handrail height above the nosing	34 to 38 in
Handrail clearance from the wall	1 1/2 in
Handrail grip — circular diameter	1 1/4 to 2 in
Handrail grip — non-circular perimeter	4 to 6 1/4 in
Handrail grip — maximum cross-section	2 1/4 in

The grip dimensions all describe the same requirement from different angles: the rail must be small enough for a hand to close right around it. A rail too fat to grip cannot be caught hold of in a fall, which is the only moment it matters.

Guards

The numbers here

Guard required when the drop exceeds	30 in
Guard height	36 in

Thirty triggers it, thirty-six satisfies it

Two numbers, one rule, and they are easy to swap. A drop of more than 30 inches means a guard is required; the guard itself must then be 36 inches tall. A guard is the barrier at the edge; a handrail is the thing you hold.

Winders

Radiating treads used where a staircase must change direction.

Stringer

The sloped member carrying the treads and risers.

Say it back

Cover the page. Answer each one out loud.

1. Maximum riser, minimum tread, and the variation allowed between them?
2. Why is the variation tolerance so tight?
3. Headroom, clear width, and maximum rise between landings?
4. Handrail height, wall clearance, and circular grip diameter?
5. At what drop is a guard required, and how tall must it be?
6. What is the difference between a guard and a handrail?
7. What are winders?